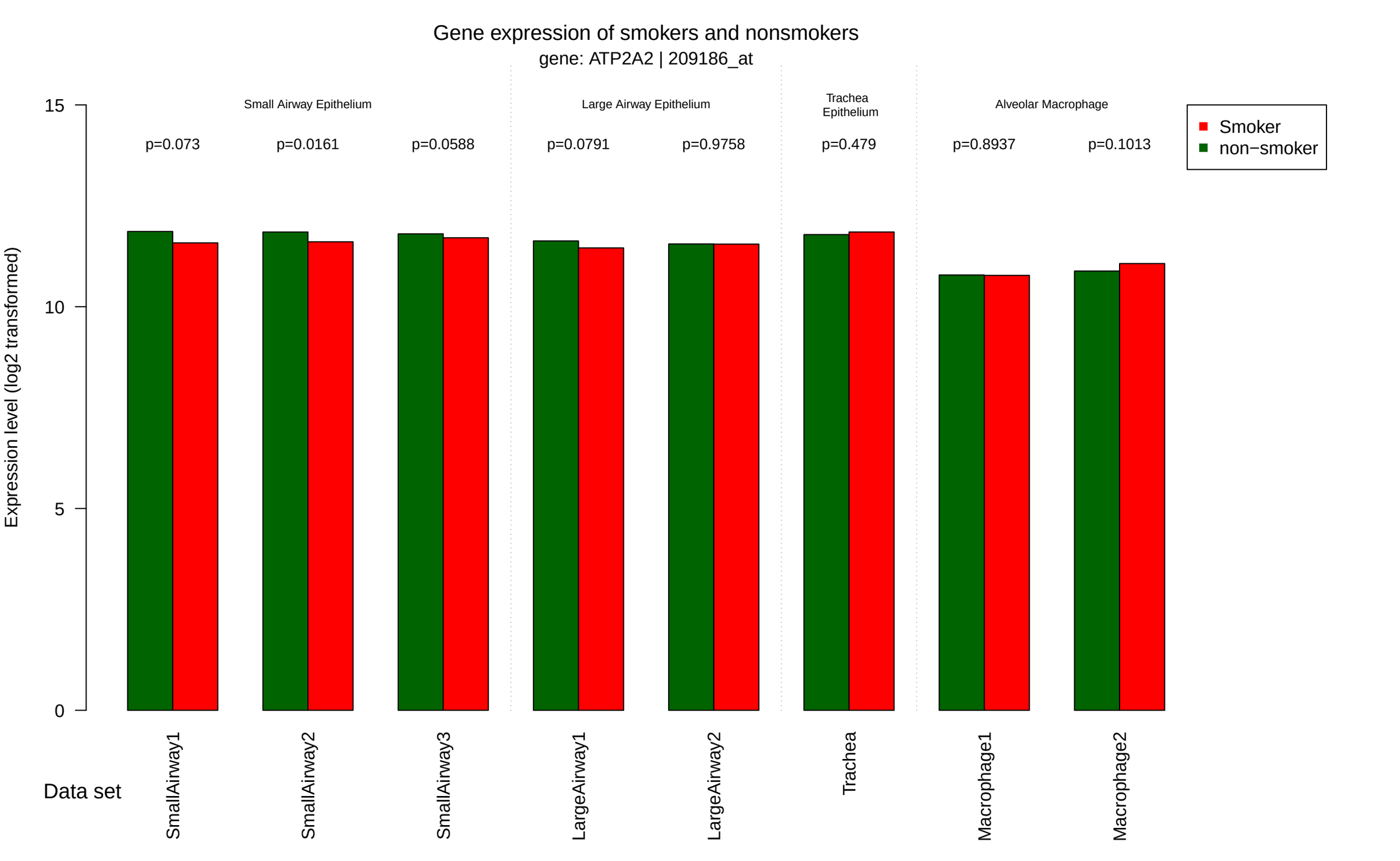
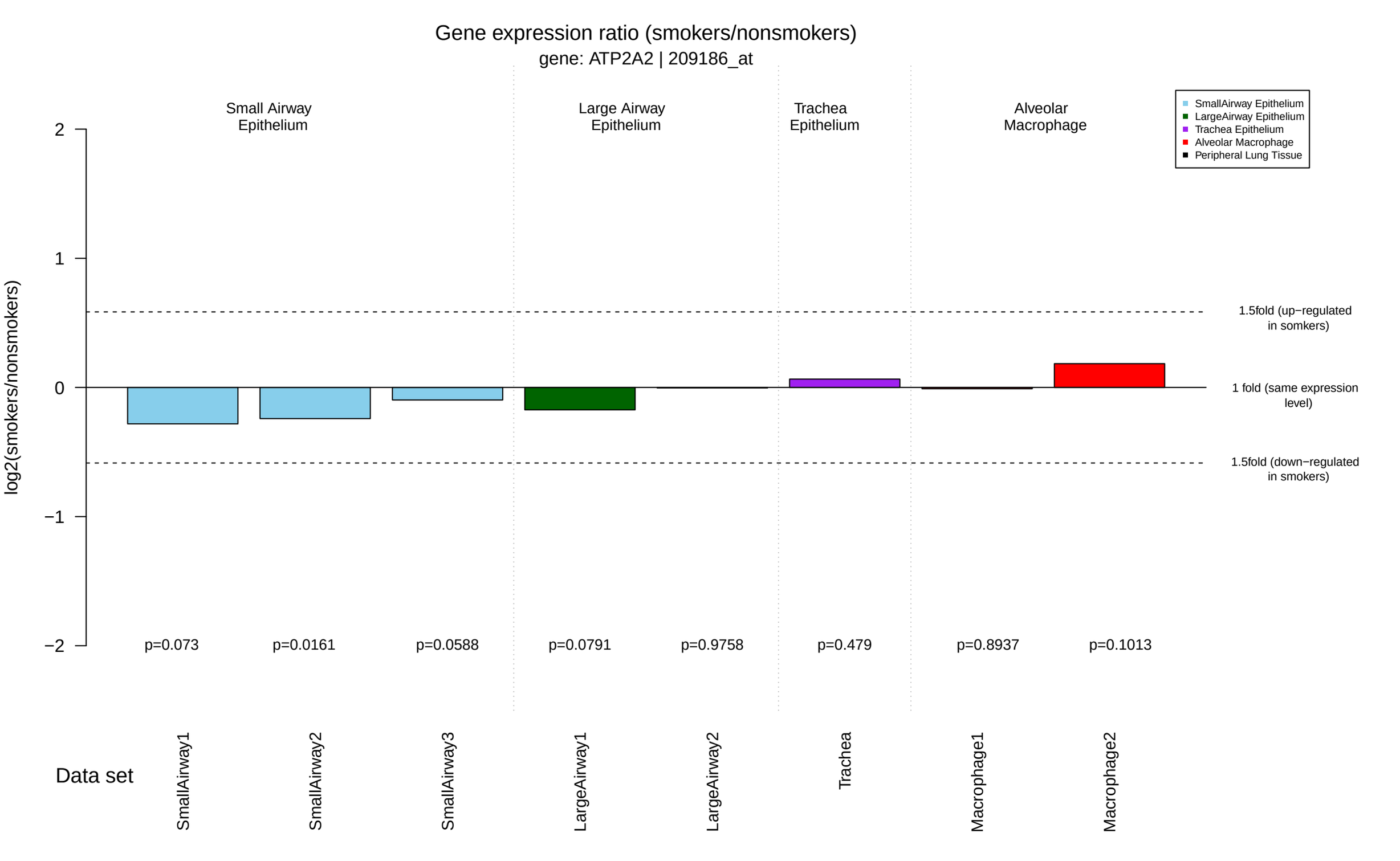
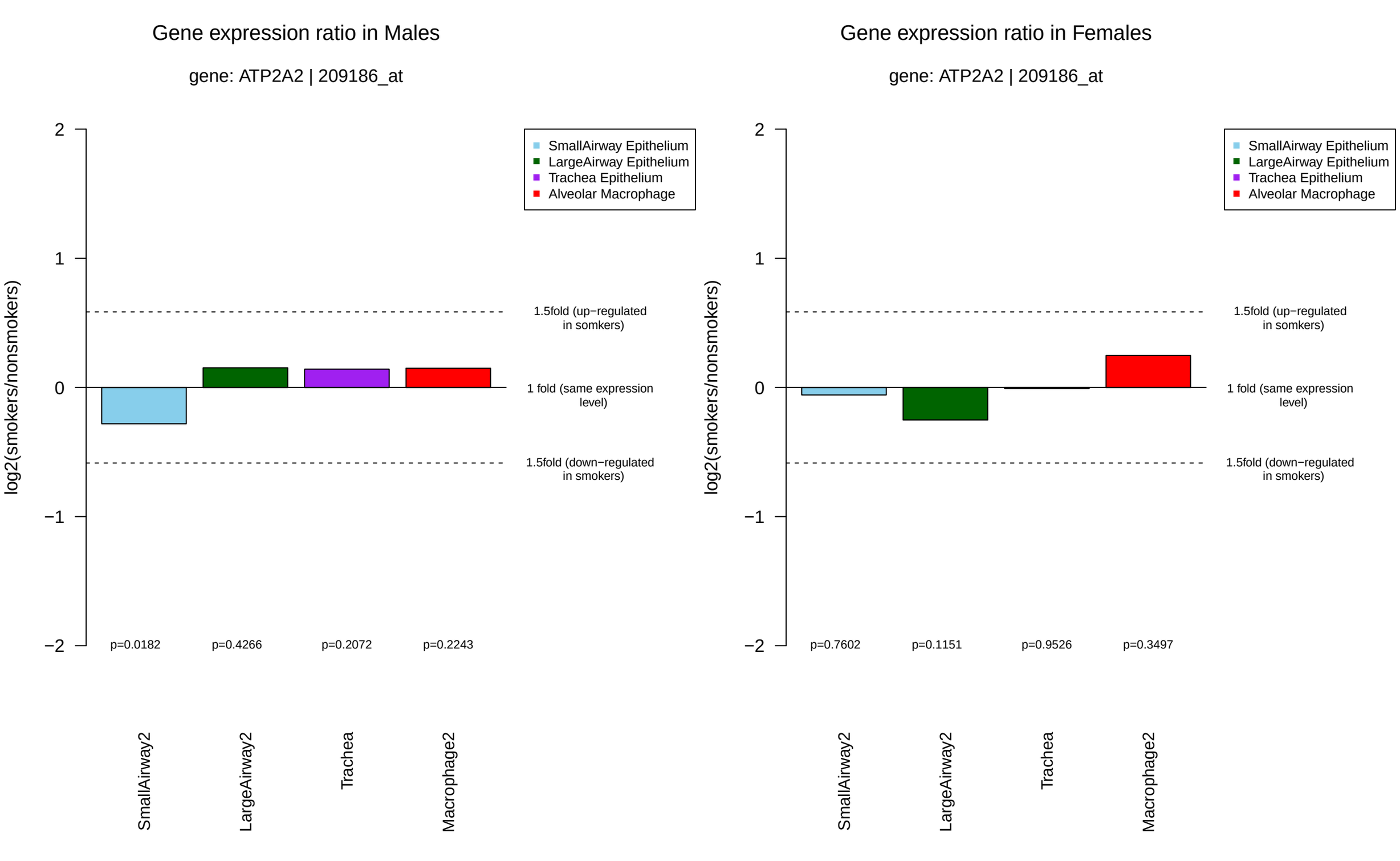
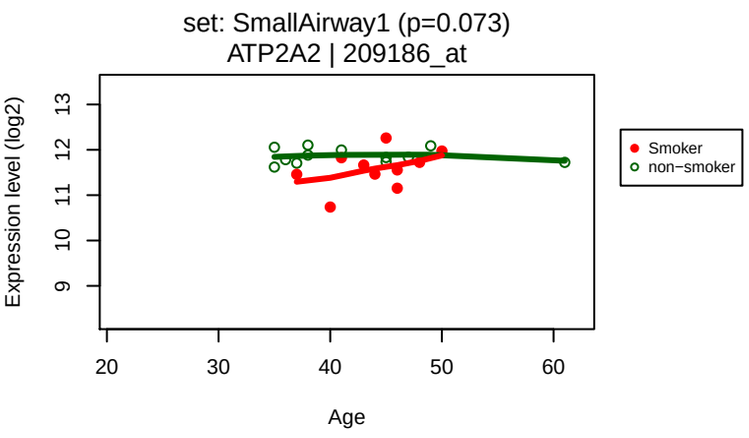
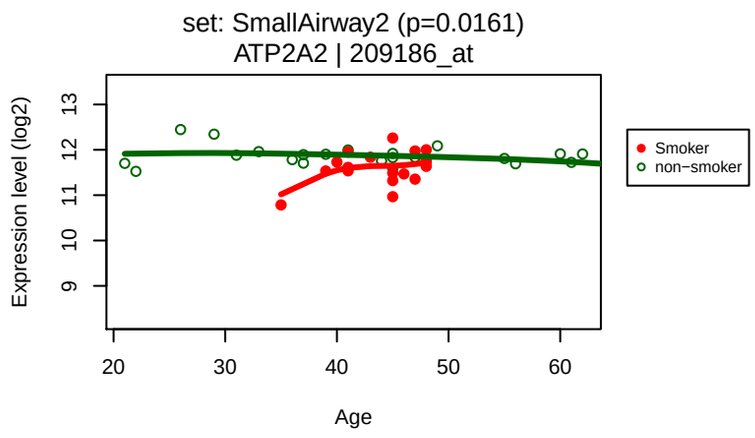
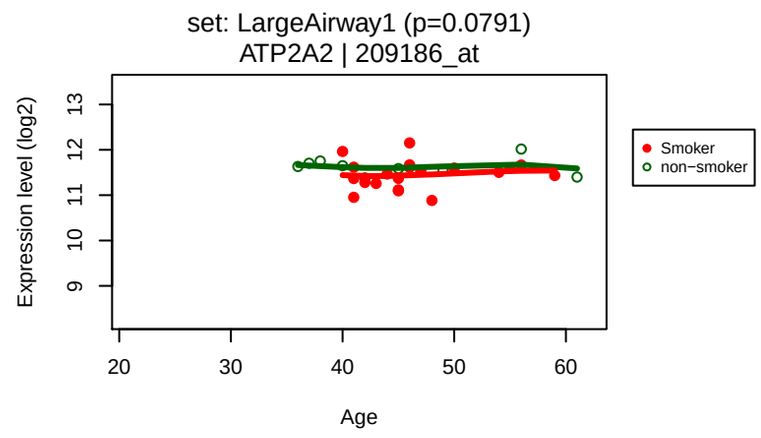

Gene expression of smokers and nonsmokers
gene: ATP2A2 | 209186_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.073
p=0.0161
p=0.0588
p=0.0791
p=0.9758
p=0.479
p=0.8937
p=0.1013
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

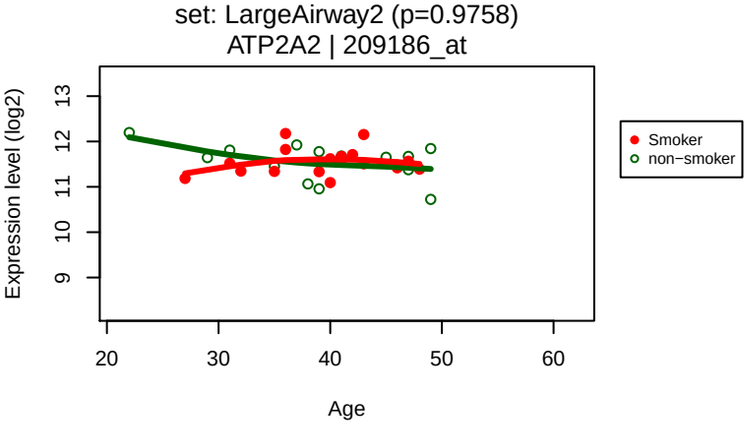

Gene expression ratio (smokers/nonsmokers)
gene: ATP2A2 | 209186_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.073
p=0.0161
p=0.0588
p=0.0791
p=0.9758
p=0.479
p=0.8937
p=0.1013
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

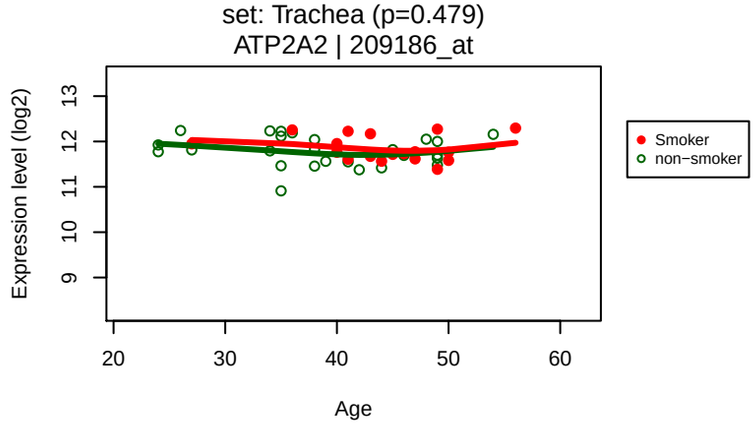

Gene expression ratio in Males
gene: ATP2A2 | 209186_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0182
p=0.4266
p=0.2072
p=0.2243
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP2A2 | 209186_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7602
p=0.1151
p=0.9526
p=0.3497
SmallAirway2
LargeAirway2
Trachea
Macrophage2

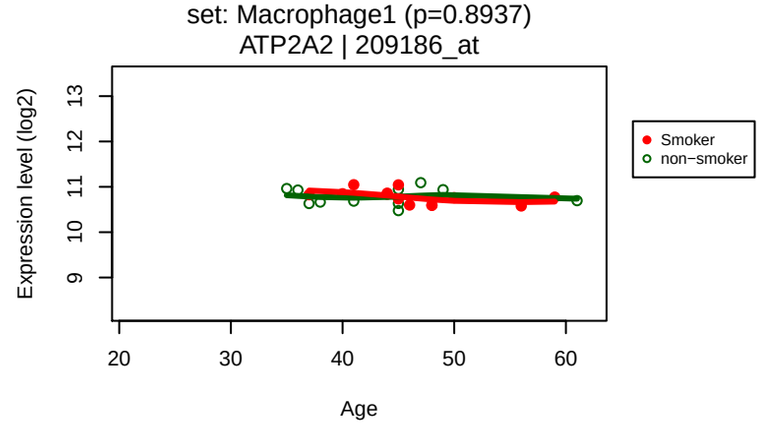

set: SmallAirway1 (p=0.073)
ATP2A2 | 209186_at
Expression level (log2)
Age
Smoker
non−smoker

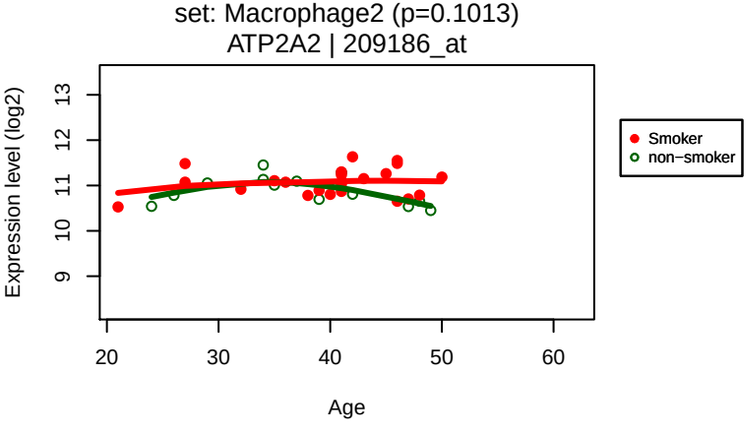

set: SmallAirway2 (p=0.0161)
ATP2A2 | 209186_at
Expression level (log2)
Age
Smoker
non−smoker

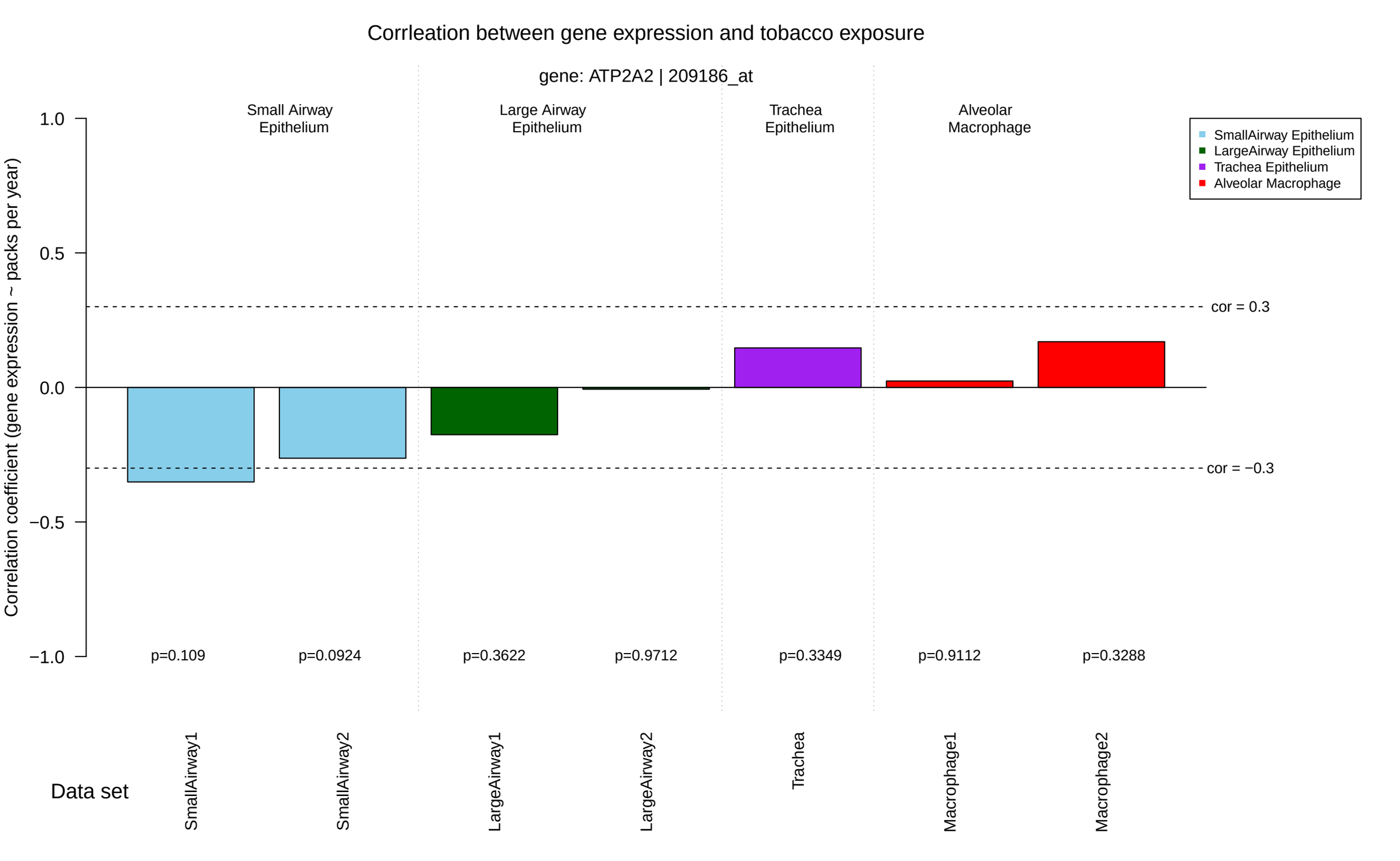

set: LargeAirway1 (p=0.0791)
ATP2A2 | 209186_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9758)
ATP2A2 | 209186_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.479)
ATP2A2 | 209186_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8937)
ATP2A2 | 209186_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1013)
ATP2A2 | 209186_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP2A2 | 209186_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.109
p=0.0924
p=0.3622
p=0.9712
p=0.3349
p=0.9112
p=0.3288
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2